| Category | ICT Moderator | N | Model | ESCS | ICT variable | ESCS × ICT | $R^2$ |
|---|---|---|---|---|---|---|---|
| | | | Baseline | 30.22*** (1.49) | — | — | 0.4040 |
| | | | Additive | 32.99*** (1.72) | -4.23*** (1.04) | — | 0.4049 |
| | ICTRES | 213,685 | Interaction | 33.74*** (2.04) | -3.83*** (1.08) | 1.31 (0.79) | 0.4053 |
| | | | Baseline | 32.09*** (1.17) | — | — | 0.3252 |
| | | | Additive | 31.95*** (1.17) | -3.21*** (0.40) | — | 0.3267 |
| | ICTSCH | 296,746 | Interaction | 31.85*** (1.17) | -2.52*** (0.46) | 1.94*** (0.36) | 0.3274 |
| | | | Baseline | 31.92*** (1.21) | — | — | 0.3194 |
| | | | Additive | 32.00*** (1.21) | -2.35*** (0.50) | — | 0.3200 |
| | ICTHOME | 287,984 | Interaction | 31.32*** (1.18) | -1.83** (0.62) | 1.09* (0.46) | 0.3201 |
| | | | Baseline | 32.04*** (1.21) | — | — | 0.3199 |
| | | | Additive | 31.66*** (1.19) | 6.99*** (0.65) | — | 0.3241 |
| Access | ICTQUAL | 287,319 | Interaction | 31.85*** (1.21) | 7.56*** (0.70) | 3.04*** (0.62) | 0.3251 |
| | | | Baseline | 32.25*** (1.20) | — | — | 0.3184 |
| | | | Additive | 32.54*** (1.19) | -13.99*** (0.81) | — | 0.3346 |
| | ICTWKEND | 277,166 | Interaction | 31.60*** (1.13) | -16.61*** (0.98) | -5.95*** (0.61) | 0.3388 |
| | | | Baseline | 32.23*** (1.21) | — | — | 0.3190 |
| | | | Additive | 32.31*** (1.19) | -16.41*** (0.76) | — | 0.3455 |
| Frequency | ICTWKDY | 282,375 | Interaction | 31.01*** (1.13) | -18.47*** (0.89) | -4.44*** (0.49) | 0.3485 |
| | | | Baseline | 32.23*** (1.24) | — | — | 0.3192 |
| | | | Additive | 32.84*** (1.24) | -9.76*** (0.84) | — | 0.3271 |
| | ICTFEED | 282,799 | Interaction | 33.42*** (1.17) | -10.51*** (0.96) | -2.93*** (0.72) | 0.3281 |
| | | | Baseline | 32.32*** (1.21) | — | — | 0.3193 |
| | | | Additive | 31.48*** (1.09) | 5.25*** (0.94) | — | 0.3215 |
| | ICTOUT | 282,352 | Interaction | 31.48*** (1.06) | 5.25*** (1.12) | 0.01 (0.98) | 0.3215 |
| | | | Baseline | 32.20*** (1.24) | — | — | 0.3198 |
| | | | Additive | 31.64*** (1.24) | 5.45*** (0.53) | — | 0.3226 |
| Use | ICTINFO | 275,677 | Interaction | 31.76*** (1.23) | 5.28*** (0.58) | -0.78 (0.47) | 0.3227 |
| | | | Baseline | 32.30*** (1.23) | — | — | 0.3221 |
| | | | Additive | 31.16*** (1.20) | 13.26*** (0.64) | — | 0.3334 |
| | INTICT | 276,693 | Interaction | 31.08*** (1.22) | 13.04*** (0.73) | -0.66 (0.55) | 0.3334 |
| | | | Baseline | 32.21*** (1.23) | — | — | 0.3205 |
| | | | Additive | 31.02 ***(1.12) | 14.04*** (0.82) | — | 0.3350 |
| Interest | DIGINTLRN | 276,810 | Interaction | 31.19*** (1.05) | 13.91*** (0.91) | -0.62 (0.60) | 0.3351 |
| | | | Baseline | 31.95*** (1.16) | — | — | 0.3284 |
| | | | Additive | 32.40*** (1.14) | -4.84*** (1.00) | — | 0.3301 |
| AI usage | AIUSESCH | 307,350 | Interaction | 32.23*** (1.16) | −5.51*** (1.14) | −3.51*** (0.88) | 0.3310 |
| * | p < .05 | All models' output is combined science scores (PV1-10SCIE), controlling for country (CNT) and gender | | | | | |
| ** | p < .01 | Estimates incorporate students and replicate weights and all ten science plausible values | | | | | |
| *** | p < .001 | ESCS and ICT variables are constructed using PISA WLE | | | | | |
| | | Coefficients are unstandardized; standard errors are shown in parentheses | | | | | |